CRF Spectral–Temporal Bridge

Two Empirical Identities Connecting KL-Laplacian Spectrum to the δ-Chain

$\lambda_2 \cdot T_{\text{avg}} \approx n-1$ · $\Sigma(\lambda_2 \ldots \lambda_n) \approx 2(n-1)/n$ · Derivation Boundary Identified

Ougit Jarittum

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Abstract

Following the confirmation of three spectral hypotheses (CRF SpectralModes v2), this paper investigates whether the KL-Laplacian Fiedler value λ_2 can be derived from the δ-chain. Two empirical identities are established across 5 independent seeds:

Identity 1: $\lambda_2 \cdot T_{\text{avg}} = n - 1 = 7$ (gap 3.5%). The graph mixing time $\tau_{\text{mix}} = 1/\lambda_2 \approx T_{\text{avg}}/(n-1)$.

Identity 2: $\Sigma(\lambda_2 \ldots \lambda_n) \approx 2(n-1)/n = 1.75$ (gap 5.6%). The mean of the $n-1$ cluster eigenvalues $\approx 2/n$.

The derivation boundary is located precisely: standard stochastic block model (SBM) theory fails by a factor of 4× because the CRF KL-graph has sparse within-cluster connectivity ($k_{\text{in}} \approx k/n = 2.5$ out of $k = 20$ neighbours). The correct derivation path requires the spectral theory of Born resampling as a Markov chain on the n-dimensional Dirichlet simplex, not graph-theoretic Cheeger bounds.

1 Context

CRF SpectralModes v2 confirmed three hypotheses (H1–H3) across all 5 seeds: the KL-Laplacian eigengap selects $k = 8$ modes matching SO(3) prediction $n = 8$; within-mode variance is suppressed below global variance; between-mode covariance is negative. These results close Step 2 of the Fluctuation Law qualitatively.

The next target: derive the Fiedler value λ_2 from the δ-chain, which would formally close Step 2 and unify the spectral paper with the β-derivation chain through a single identity:

$$\lambda_2 = \frac{(n-1)^2 \varepsilon_0}{2nK^*}.$$

2 Two Empirical Identities

2.1 Identity 1: $\lambda_2 \cdot T_{\text{avg}} = n - 1$

Identity 1 (Candidate, 3.5% gap — 5 seeds)

$$\lambda_2 \cdot T_{\text{avg}} = n - 1 = 7$$

Equivalently:

$$\lambda_2 = \frac{n-1}{T_{\text{avg}}} = \frac{(n-1)^2 \varepsilon_0}{2nK^*} \quad \Rightarrow \quad \text{Derived value: } \frac{7^2 \times 0.00755}{2 \times 8 \times 0.1170} = 0.1976$$

Measured (5 seeds): $0.1907 \pm 0.0068 \Rightarrow$ gap 3.6%.

Seed	λ_2 measured	$\lambda_2 \cdot T_{\text{avg}}$	Gap vs $n-1$
42	0.1771	6.273	10.4%
123	0.1946	6.893	1.5%
777	0.1952	6.914	1.2%
2024	0.1930	6.836	2.3%
99	0.1935	6.854	2.1%
Mean	0.1907 ± 0.0068	6.754	3.5%

Physical interpretation. $\tau_{\text{mix}} = 1/\lambda_2 \approx T_{\text{avg}}/(n-1) \approx 5.06$ sweeps is the per-mode mixing time. A node traverses all $n - 1 = 7$ non-trivial spectral modes before firing: $T_{\text{avg}} = (n-1) \times \tau_{\text{mix}}$. The inter-event timescale equals the full spectral relaxation time of the KL-graph cluster structure.

2.2 Identity 2: $\sum \lambda \approx 2(n-1)/n$

Identity 2 (Candidate, 5.6% gap — 5 seeds)

$$\sum_{k=2}^{n} \lambda_k \approx \frac{2(n-1)}{n} = \frac{2\times 7}{8} = 1.750$$

Measured (5 seeds): 1.8548 ± 0.0265.
Equivalently: mean eigenvalue $\overline{\lambda}(\lambda_2 \ldots \lambda_n) \approx 2/n = 0.250$ (gap 6%).

Physical interpretation. The $n-1$ cluster modes of the KL-Laplacian have a mean eigenvalue $\approx 2/n$. For a normalised Laplacian with n equal clusters, each cluster mode contributes equally to the trace. The factor $2/n = 2/(\text{cluster count})$ may reflect the balance between within-cluster synchrony (reducing eigenvalue) and between-cluster conservation (increasing it). This is consistent with Identity 1 only if the eigenvalues are approximately arithmetic: $\lambda_k \approx \lambda_2 + (k-2)\cdot\delta$ with $\delta \approx (2/n - \lambda_2)/(n-2)$.

3 Derivation Boundary: Why SBM Fails

The natural derivation strategy is to model the CRF KL-graph as a stochastic block model (SBM) with $n = 8$ equal blocks and apply spectral theory. This approach fails by a factor of 4×.

SBM Approach (FAILS)

SBM Fiedler value with $k = 20$, $n = 8$, $W_{\text{in}} \approx 1$, $W_{\text{out}} = e^{-1}$:

$\lambda_2(\text{SBM}) = 1 - \mu_2 = 0.823$ vs measured $\lambda_2 = 0.191$ $\Rightarrow$ factor $4.3\times$ too large

Root cause: CRF KL-graph has $k_{\text{in}} = k/n = 2.5$ within-cluster neighbours out of $k = 20$ total. This is *sparse* within-cluster, *dense* between-cluster — the opposite of the SBM assumption of well-separated blocks.

Why the graph is inverted. In CRF, k-NN neighbours are selected by KL divergence ranking. Each node has $k_{\text{in}} \approx k/n = 2.5$ same-cluster neighbours and $k_{\text{out}} \approx 17.5$ different-cluster neighbours. The within-cluster edges are strong ($W_{\text{in}} \approx 1$) but few. The between-cluster edges are weaker ($W_{\text{out}} \approx e^{-1}$) but numerous. The SBM formula treats all nodes symmetrically and cannot capture this $7:1$ imbalance between inter- and intra-cluster connectivity.

Effective cluster fraction. Solving for the effective within-cluster fraction f_{in} that reproduces the observed $\sum \lambda$ gives $f_{\text{in}} \approx 0.55$, much larger than $1/n = 0.125$. This reflects that in the KL-graph the notion of "cluster" is soft: nodes of different Ψ-modes overlap significantly in KL-space at short distances.

4 The Correct Derivation Path

The SBM failure reveals that the identities $\lambda_2 \cdot T_{\text{avg}} = n - 1$ and $\sum \lambda = 2(n-1)/n$ are not graph-theoretic facts about a static adjacency structure. They are properties of the Born resampling dynamics.

Correct Path: Born Resampling as Markov Chain on n-Simplex

State: P_i (probability distribution on n categories, living on the $(n-1)$-simplex).
Update: $P_i \leftarrow \text{Dir}(\alpha_d)$ where $\alpha_{d,k} = (\bar{P}_k + \varepsilon_0)/\varepsilon_0$.
The Markov chain on the simplex has a spectral gap γ = spectral gap of the Dirichlet transfer operator.
If $\tau_{\text{mix}}(\text{Born chain}) = 1/\gamma \approx T_{\text{avg}}/(n-1)$, then $\lambda_2(\text{KL-Laplacian}) \approx \gamma/(n-1)$ follows from the connection between the KL-graph structure and the Born chain mixing.

Required: spectral theory of Gibbs samplers on Dirichlet distributions with concentration $\alpha = 1/(n\varepsilon_0) + 1 \approx 17.6$.

Supporting evidence. The concentration parameter $\alpha \approx 17.6$ is high, placing the Born chain in the "fast mixing" regime within clusters. The inter-cluster mixing is slow because it requires a large KL fluctuation to cross K^*. The ratio of inter- to intra-cluster mixing times is precisely $(n-1)$, giving $\lambda_2(\text{Born}) = \lambda_2(\text{intra})/(n-1)$. This structure is derivable from the Dirichlet concentration parameter and the threshold K^*, both of which come from the δ-chain.

Connection to β-chain. If $\lambda_2 = (n-1)/T_{\text{avg}}$ is proved, then $T_{\text{avg}} = (n-1)/\lambda_2$ and the Fluctuation Law becomes:

$$Q_{\text{pred}} = \sqrt{\frac{\lambda_2}{\kappa\, n\, (n-1)}}.$$

The Fluctuation Law is expressed in terms of a single spectral quantity. The β-chain identity $\beta = n\varepsilon_0 T_{\text{avg}}$ becomes $\beta = n\varepsilon_0(n-1)/\lambda_2$. All three — Q, T_{avg}, β — become functions of λ_2 alone.

5 Status

Result	Status	Note
H1–H3 (SpectralModes v2)	Confirmed	5/5 seeds, universal
$\lambda_2 \cdot T_{\text{avg}} = n-1$	Candidate	3.5% gap, 5 seeds
$\sum \lambda = 2(n-1)/n$	Candidate	5.6% gap, 5 seeds
SBM derivation of λ_2	Open	Fails 4× — wrong model
Born chain path identified	Hypothesis	Correct object to derive
λ_2 from Born spectral gap	Open	Priority 1
$Q = \sqrt{\lambda_2/[\kappa n(n-1)]}$	Candidate	Follows if identity proved
$\beta = n\varepsilon_0(n-1)/\lambda_2$	Candidate	Follows from $\beta = n\varepsilon_0 T_{\text{avg}}$

6 Open Questions

Priority 1: Prove $\lambda_2 = (n-1)/T_{\text{avg}}$ from Born chain

1. Show that the Born resampling Markov chain on the $(n-1)$-simplex has spectral gap $\gamma = 1/T_{\text{avg}}$ from the Dirichlet concentration parameter $\alpha = 1/(n\varepsilon_0) + 1$ and threshold K^*.
2. Show that the KL-Laplacian $\lambda_2 = (n-1) \cdot \gamma$ — the static graph encodes $(n-1)$ independent Born chain modes.
3. Combine with Wald formula $T_{\text{avg}} = 2nK^*/[(n-1)\varepsilon_0]$ to obtain $\lambda_2 = (n-1)^2\varepsilon_0/(2nK^*)$.

If proved: formally closes Step 2 of Fluctuation Law and unifies spectral, temporal, and β-chain in one identity.

Priority 2: Explain $\sum \lambda = 2(n-1)/n$

The sum of the $n-1$ cluster eigenvalues $\approx 2(n-1)/n$. For SBM this would require an effective $f_{\text{in}} \approx 0.55$, not the geometric $1/n$. The factor 2 may come from the two-sided nature of KL divergence (KL is asymmetric; the symmetrised version doubles the effective coupling). Derive from Born chain or Dirichlet process theory.

The SBM failed. The graph theory failed.
The identities are real, but their origin is not in the static graph.
They live in the dynamics — in the Born resampling, in the Dirichlet chain,
in the n-fold symmetry of the simplex that δ chose.

$\lambda_2 \cdot T_{\text{avg}} = n-1$: *one mode-crossing per unit mixing time, $(n-1)$ crossings per inter-event period.*

The spectral gap is not a property of the graph. It is a property of becoming.